MA2214

AY25/26 Sem 1

by ngmh

Basic Counting Principles

Listing List every object, then establish a bijection between it and $[n] = \{1, ..., n\}$ which has n elements.

Addition Principle (AP) For events E_k with n_k ways to occur, if they are pairwise disjoint, the number of ways where at least one event occurs is $\sum_{i=1}^{k} n_i$, and if $E = E_1 \sqcup ... \sqcup E_r$ then $|E| = \sum_{i=1}^{r} n_i$.

Multiplication Principle (MP) For an event E which can be decomposed into r ordered events $E_1, ..., E_k$, where E_k has n_k ways to occur, then there are $\prod_{i=1}^{k} n_i$ ways for E to occur, and if $E = E_1 \times ... \times E_r$, then $|E| = n_1 ... n_r$.

r-Permutation An r-permutation of A is a way of arranging any r of the objects of A in a row, with number denoted as $P_r^n = \frac{n!}{(n-r)!}$ by MP.

Principle of Complementation (CP) If $A \subseteq$ a finite universal set U, then $|U \backslash A| = |U| - |A|$.

Principle of Division If we can partition instances of an event E into disjoint r-element groups which each gives rise to a unique E', then $|E'| = \frac{|E|}{r}$.

Circular Permutation An r-circular permutation of A is a circular permutation of r distinct objects taken from A, denoted $Q_r^n = \frac{P_r^n}{r}$.

Combination An r-combination of A is an r-element subset of A, denoted by $C_r^n = \binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$. Additionally, $\binom{n}{r} = \binom{n}{n-r}$.

Combinatorial Identities

- $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$ (Either take the current element, or choose r from the set without the current element)
- $\binom{n}{r} = \frac{n}{r}\binom{n-1}{r-1}$ (Move r over, then LHS becomes choosing r then a leader while RHS becomes choosing the leader then remaining $r-1$)
- $\binom{n}{r} = \frac{n-r+1}{r}\binom{n}{r-1}$
- $\binom{n}{r} = \frac{n}{n-r}\binom{n-1}{r}$
- $\binom{n}{m}\binom{m}{r} = \binom{n}{r}\binom{n-r}{m-r}$

Principle of Inclusion and Exclusion $|A_1 \cup A_2 \cup ... \cup A_q| = \sum_{i=1}^{q} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| ... + (-1)^{q-1}|A_1 \cap ... \cap A_q|$

Euler's φ Function $\varphi(n)$ denotes the number of integers between 1 and n which are coprime to n, and $\varphi(n) = n(1 - \frac{1}{p_1})...(1 - \frac{1}{p_k}) = n\prod_{i=1}^{k}(1 - \frac{1}{p_i})$

Injection Principle If there is a injection from A to B, then $|A| \leq |B|$

Surjective Principle If there is a surjection from A to B, then $|A| \geq |B|$

Bijection Principle If there is a bijection from A to B, then $|A| = |B|$

Twelvefold Way

Objects / Boxes	Arbitrary	Injective	Surjective
Distinct / Distinct	n^r	$P(n,r) = \frac{n!}{(n-r)!}$	$n!\,S(r,n)$
Distinct / Identical	$\sum_{k=0}^{n} S(r,k)$	$\begin{cases} 1 & r \leq n \\ 0 & r > n \end{cases}$	$S(r,n)$
Identical / Distinct	$\binom{r+n-1}{r}$	$\binom{n}{r}$	$\binom{r-1}{n-1}$
Identical / Identical	$\sum_{k=0}^{n} p(r,k)$	$\begin{cases} 1 & r \leq n \\ 0 & r > n \end{cases}$	$p(r,n)$

- Injective: Each box has at most one object
- Surjective: Each box has at least one object
- $S(r,n)$: Stirling numbers of the second kind
- $p(r,n)$: Number of partitions of r identical objects into exactly n identical boxes

Non-Negative Integer Solutions For the linear equation $x_1 + ... + x_n = r$ where $x_i \geq 0$, the number of solutions is $\binom{r+n-1}{r}$ using stars and bars.

Multiset $M = \{r_1 \cdot a_1, r_2 \cdot a_2, ..., r_n \cdot a_n\}$ where r_i is the repetition number of object a_i. $\infty \cdot a$ is used to indicate infinite copies of an object.

Multiset Permutations The number of r-permutations of $\{\infty \cdot a_1, ..., \infty \cdot a_n\}$ is n^r. If $r = \sum_1^n r_i$, the number of permutations of M, $P(r; r_1, ..., r_n) = \frac{r!}{r_1! ... r_n!}$

Combinations with Repetition A m element multi-subset of M is a subset $\{m_1 \cdot a_1, ..., m_n \cdot a_n\}$ where $\sum_1^n m_i = m$. The number of r-element multi-subsets is given by $\binom{r+n-1}{r}$.

Stirling Numbers of the Second Kind $S(r,n)$ or $\left\{ {r \atop n} \right\}$ counts the number of ways to partition a set of r distinct objects into n **non-empty**, unlabeled subsets. Base Cases: $S(0,0) = 1, S(r,0) = 0, r > 0, S(0,n) = 0, n > 0, S(r,n) = 0, r < n$. Recursive Definition: $S(r,n) = nS(r-1,n) + S(r-1,n-1)$; either the rth element is in its own subset or it joins an existing subset.

Explicit formula for Stirling Numbers of the Second Kind $S(r,n) = \frac{1}{n!}\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}j^r$. Derived from $n! \cdot S(r,n) = \sum_{j=0}^{n}(-1)^j\binom{n}{j}(n-j)^r$, where LHS is number of surjective functions from r elements to n elements, while RHS is the same by considering PIE on the set of functions avoiding element i. $S(n,k) \sim \frac{k^n}{k!}$ as $n \to \infty$

Partitions of an Integer A partition of r is writing r as a sum of positive integers where order does not matter. The number of partitions of r is $p(r)$, and $p(r,n)$ if into n sums.

- $p(0,0) = 1, p(r,0) = 0, p(0,n) = 0$
- $p(r,n) = p(r-1,n-1) + p(r-n,n)$: Either one summand is 1, or all of them are at least 2 so we can subtract 1 from all.

Ferrers Diagram Graphical representation of integer partition as rows of dots or boxes, left-aligned, ordered in non-increasing order.

Conjugate Partition To get a conjugate partition, we can simply transpose it. The number of partitions of r into n parts is thus the same as the number of partitions of n into r parts.

Uses of Partitions We can use partitions to analyse time complexity especially when the algorithm involves subdivisions.

Polynomials

- A polynomial with coefficients in A and variable x is an expression $P(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0$ where n is non-negative, $a_i \in A, a_n \neq 0$. $\mathbb{Z}[x]$ is the collection of polynomials where $A = \mathbb{Z}$.
- Polynomials for a commutative ring under addition and multiplication.
- To evaluate a polynomial at a constant c, sub $x = c$. Evaluation preserves the ring structure of polynomials.

Sequences to Polynomials For a finite sequence $a_0, a_1, ..., a_n$, we associate it with the polynomial $P(x) = a_0 + a_1 x + ... + a_n x^n$. This is linear and bijective.

Binomial Coefficients $\binom{n}{r} = \frac{n!}{r!(n-r)!}$'s are known as the binomial coefficients.

Binomial Theorem $\forall n \geq 0, (x+y)^n = \sum_{r=0}^{n}\binom{n}{r}x^{n-r}y^r$. Consider the string of length n with letters x or y. This can be used to prove identities by subbing in appropriate values for x, y.

Example Identities $\sum_{r=0}^{n}(-1)^r\binom{n}{r} = 0$ using $(1 + (-1))^n$, $\binom{n}{0} + \binom{n}{2} + ... + \binom{n}{2k} = \binom{n}{1} + \binom{n}{3} + ... + \binom{n}{2k+1} = 2^{n-1}$

Another Identity $\sum_{r=1}^{n} r\binom{n}{r} = n \cdot 2^{n-1}$ by differentiating $(1+x)^n$ and subbing in $x = 1$

Multinomial Coefficients Consider $\binom{n}{n_1, n_2, ..., n_m}$ where their sum is n. Then $\binom{n}{n_1, n_2, ..., n_m} = \binom{n}{n_1}\binom{n-n_1}{n_2} ... = \frac{n!}{n_1! n_2! ... n_m!}$.

Multinomial Theorem For $n, m \in \mathbb{N}, (x_1 + x_2 + ... + x_m)^n = \sum \binom{n}{n_1, n_2, ..., n_m} x_1^{n_1} x_2^{n_2} ... x_m^{n_m}$ where we consider all m-ary sequences of nonnegative integers summing to n.

Ordinary Generating Functions

Power Series

- A power series with coefficients in A and variable x is an expression of the form $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + ...$ where $a_n \in A$. $\mathbb{Z}[[x]]$ is the collection of power series where $A = \mathbb{Z}$.
- Addition is performed coefficient-wise. Multiplication is defined by the Cauchy product $(\sum_{n=0}^{\infty} a_n x^n)(\sum_{n=0}^{\infty} b_n x^n) = \sum_{n=0}^{\infty}(\sum_{k=0}^{n} a_k b_{n-k})x^n$. Power series form a commutative ring over $\mathbb{R}$ or $\mathbb{C}$ under addition and multiplication.
- To evaluate a power series $F(x) = \sum_{n=0}^{\infty} a_n x^n$ at a constant c, substitute $x = c$ where $|c| < R$, the radius of convergence. Evaluation is only valid inside the radius of convergence and preserves the ring structure of power series.

Sequences to Power Series For an infinite sequence $a_0, a_1, a_2, ...$, it is associated with the power series $F(x) = a_0 + a_1 x + a_2 x^2 + ... = \sum_{n=0}^{\infty} a_n x^n$. This is linear and bijective, and $F(x)$ is the generating function for the sequence.

Cauchy Product of Series The product of two series $F(x)G(x)$ corresponds to the sequence $c = (c_0, c_1, c_2, ...), c_r = \sum_{k=0}^{r} a_k b_{r-k}$. For multiple series, the product corresponds to $c_r = \sum_{k_1 + k_2 + ... + k_m = r} a_{k_1}^{(1)} a_{k_2}^{(2)} ... a_{k_m}^{(m)}$.

Encoding Sets For $A^{(i)} \subseteq \mathbb{Z}_{\geq 0}$, define $F^{(i)}(x) = \sum_{k \in A^{(i)}} x^k$. Let $F(x) = F^{(1)}(x) \cdot ... \cdot F^{(m)}(x)$. Then the coefficient c_r of x^r in $F(x)$ counts solutions of $x_1 + x_2 + ... + x_m = r, x_i \in A^{(i)}$.

Weighted Sums To count solutions of $x_1 + 2x_2 + ... + mx_m = r$, we can use generating functions with weights in exponents. For x_i with coefficient i, associate $F^{(i)}(x) = \sum_{k \in A^{(i)}} x^{ik}$. Then $F(x) = F^{(1)}(x) ... F^{(m)}(x)$ and the coefficient c_r of x^r counts solutions.

Cumulative Sums Let $F(x) = \sum_{n=0}^{\infty} a_n x^n$. Then $(1 + x + x^2 + ...) \cdot F(x) = (a_0) + (a_0 + a_1)x + ...$ encodes $c_r = a_0 + ... + a_r$.

Constrained Equation $x_1 \geq 1 \Rightarrow F_1(x) = x + x^2 + x^3 + ..., x_2 \leq 2 \Rightarrow F_2(x) = 1 + x + x^2$. The genfunc is $F(x) = F_1(x)F_2(x)...$. Then we can use coefficients to find the number of solutions. Useful tip: For $(1 + x + ...)^2$, the coefficient of x^k is $k+1$

Dividing Genfuncs Division is costly, use partial fraction decomposition instead, e.g. $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \frac{1}{1-2x} = \sum_{n=0}^{\infty} 2^n x^n$.

Partition Genfunc The genfunc for number of partitions $p(n)$ is $\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty}(1 + x^k + x^{2k} + ...) = \prod_{k=1}^{\infty}\frac{1}{1-x^k}$.

Euler's Partition Theorem Again $D(x) = \prod_{k=1}^{\infty}(1 + x^k)$, $O(x) = \prod_{k=1}^{\infty}\frac{1}{1-x^{2k-1}}$. Using $1 + x^k = \frac{1-x^{2k}}{1-x^k}$, we get what we want.

Exponential Generating Functions The EGF for a sequence (a_r) is $a_0 + a_1\frac{x}{1!} + ... + a_r\frac{x^r}{r!} + ... = \sum_{r=0}^{\infty} a_r\frac{x^r}{r!}$. $a_r = r!c_r$ where c_r is the rth coefficient of the genfunc.

OGF v/s EGF OGF are applicable in distribution or arrangement problems where ordering of objects involved is immaterial. EGF are useful in counting arrangements of objects where ordering is taken into account.

General EGF Formula Let a_r denote the number of r-permutations of the multi-set $\{n_1 \cdot b_1, ..., n_k \cdot b_k\}$. Then the EGF is $(1 + \frac{x}{1!} + ... + \frac{x_{b_1}^{n_1}}{n_1!})...(1 + \frac{x}{1!} + ... + \frac{x_{b_k}^{n_k}}{n_k!})$.

Recurrence Relations

Relating Indexes

- $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$
- $\left[{n+1 \atop k}\right] = n\left[{n \atop k}\right] + \left[{n \atop k-1}\right]$ (Stirling number of the first type, permutations of n elements with k disjoint cycles)
- $\left\{{n+1 \atop k}\right\} = k\left\{{n \atop k}\right\} + \left\{{n \atop k-1}\right\}$ (Stirling number of the second type, partition of n elements into k nonempty subsets)

Recurrence Relation Relates current term to some previous terms, with base cases known as initial conditions.

Difference Equation Write in the form $\Delta a_n = a_{n+1} - a_n$.

Linear Homogeneous Recurrence Relations Recurrence relation with the form $c_0a_n + c_1a_{n-1} + ... + c_ra_{n-r} = 0$. A function with $a_n = f(n)$ is a solution of the recurrence relation.

Characteristic Equation For a linear homogeneous recurrence relation, replace a_i by x^i, then divide by x^{n-r} to get the characteristic equation. Any root of this equation is a characteristic root.

LHRR Theorem If $\alpha_1, ..., \alpha_r$ are the distinct characteristic roots of our RR, then $a_n = A_1(\alpha_1)^n + ... + A_r(\alpha_r)^n$ where A_i is constant are general form of solutions to our RR. We can sub in the initial condition after we find the general form to find the suitable solution.

Complex Roots The solution to a LHRR might have complex roots which can be expressed using $e^{ix} = cos\ x + isin\ x$.

LHRR Theorem Again If $\alpha_1, ..., \alpha_k$ are the distinct characteristic roots such that α_i has multiplicity m_i, the general form of the solution of our recurrence relation is $a_n = (A_{11} + A_{12}n + ... + A_{1m_1}n^{m_1-1})(\alpha_1)^n + ... + (A_{k1} + A_{k2}n + ... + A_{km_k}n^{m_k-1})(\alpha_k)^n$.

General Linear Recurrence Relation A recurrence relation of the form $c_0a_n + c_1a_{n-1} + ... + c_ra_{n-r} = f(n)$ is an rth order linear recurrence relation. An LHRR is a linear recurrence relation where $f(n) = 0$. If $g_1(n), g_2(n)$ are solutions for above, then $h(n) = g_1(n) - g_2(n)$ is a solution of the corresponding LHRR.

Solving General Linear Recurrence Relations

- Find the general solution $h(n)$ of the corresponding LHRR.
- Find a particular solution $g_0(n)$ of the linear recurrence relation. It can be chosen as a function of a similar type.
- The general family of solutions of $g_0(n)$ of the original recurrence relation is given by $g(n) = g_0(n) + h(n)$.
- Find the specific solution that fits the given initial conditions.
- For a polynomial of degree d, try a polynomial of degree d (or higher if the homogeneous solution contains polynomial terms)
- For an exponential function k^n, try $C \cdot k^n$ (or $C \cdot n^m \cdot k^n$ if k is a root of the characteristic equation with multiplicity m)
- For a combination of these forms, try a combination of the corresponding trial solutions

Example Recurrence Relation

- Solve $a_n - 3a_{n-1} + 2a_{n-2} = 2^n$ with $a_0 = 3, a_1 = 8$
- First solve the LHRR $a_n^{(h)} - 3a_{n-1}^{(h)} + 2a_{n-2}^{(h)} = 0$
- The characteristic equation is $r^2 - 3r + 2 = 0, r = 1, 2$
- We get $a_n^{(h)} = A \cdot 1^n + B \cdot 2^n = A + B \cdot 2^n$
- Now find a particular solution by trying $a_n^{(p)} = C \cdot n \cdot 2^n$ since $f(n) = 2^n$ and 2 is a root of the characteristic equation
- Substituting gives us $Cn2^n - 3C(n-1)2^{n-1} + 2C(n-2)2^{n-2} = 2^n$ and solving gives $C = 2$
- $a_n^{(p)} = 2n \cdot 2^n = n \cdot 2^{n+1}$
- The general solution is $a_n = a_n^{(h)} + a_n^{(p)} = A + B \cdot 2^n + n \cdot 2^{n+1}$
- Applying initial conditions gives $A = 2, B = 1$
- The final solution is $a_n = 2 + (2n+1)2^n$

$g_0(n)$ Chart

- Ak^n, k is not characteristic root: Bk^n
- Ak^n, k is characteristic root with multiplicity m: Bn^mk^n
- $\sum_{i=0}^t p_in^i$, 1 is not characteristic root: $\sum_{i=0}^t q_in^i$
- $\sum_{i=0}^t p_in^i$, 1 is a characteristic root of multiplicity m: $n^m\sum_{i=0}^t q_in^i$
- An^tk^n, k is not a characteristic root: $(\sum_{i=0}^t q_in^i)k^n$
- An^tk^n, k is a characteristic root of multiplicity m: $n^m(\sum_{i=0}^t q_in^i)k^n$

Solving Linear Recurrence Relations with GF

- Transform equation into linear equation for generating function
- Solve linear equation and simplify the answer, often involving partial fractions
- Fibonacci: $a_0 = 0, a_1 = 1, a_n = a_{n-1} + a_{n-2}$
- Define $A(x) = \sum_{n=0}^\infty a_nx^n$.
- Then $A(x) - x - 0 = \sum_2^\infty a_nx^n = \sum_2^\infty(a_{n-1} + a_{n-2})x^n = x\sum_1^\infty a_nx^n + x^2\sum_0^\infty a_nx^n = x \cdot A(x) + x^2 \cdot A(x)$.
- We have $A(x) - x = x \cdot A(x) + x^2 \cdot A(x) \Rightarrow A(x) = \frac{x}{1-x-x^2}$.
- We have $1 - x - x^2 = (1-r_1x)(1-r_2x)$, and $A(x) = \frac{1}{\sqrt{5}}(\frac{1}{1-r_1x} - \frac{1}{1-r_2x})$. Then we get $a_n = \frac{1}{\sqrt{5}}(r_1^n - r_2^n)$ by comparing coefficients.

Partial Fractions

- Distinct Linear Factors: $\frac{P(x)}{(1-r_1x)(1-r_2x)...(1-r_kx)} = \sum_{i=1}^k \frac{A_i}{1-r_ix}$
- Each term expands as $\frac{1}{1-rx} = \sum_{n=0}^\infty r^nx^n$
- Repeated Roots: If a factor has multiplicity m: $\frac{P(x)}{(1-rx)^m} = \frac{B_1}{1-rx} + ... + \frac{B_m}{(1-rx)^m}$
- Expansion gives terms of the form $\frac{1}{(1-rx)^j} = \sum_{n=0}^\infty \binom{n+j-1}{j-1}r^nx^n$
- Use $x = \frac{1}{r_i}$ to eliminate and solve for other coefficients

System of Linear Recurrence Relations

- Domino Tiling: Tile $2 \times n$ board using 1×2 and 1×1 tiles
- Let a_n be complete tilings, and b_n be tilings with a missing square
- $a_n = b_n + a_{n-1} + b_{n-1} + a_{n-2}$
- $b_n = a_{n-1} + b_{n-1}$
- Use substitution to eliminate one sequence, then spam genfunc

Tutorial Stuff

More Identities

- $P_r^n = nP_{r-1}^{n-1}$
- $P_r^n = (n-r+1)P_{r-1}^n$
- $P_r^n = \frac{n}{n-r}P_r^{n-1}$
- $P_r^{n+1} = P_r^n + rP_{r-1}^n$
- $P_r^{n+1} = r! + r(P_{r-1}^n + P_{r-1}^{n-1} + ... + P_{r-1}^r)$: Either take the last r elements, or choose where to place $n+1$, then don't take $n+1, n, n-1, ..., i+1$, then take i

Divisibility Argument

- Prove $(n+1)(n+2)...(2n)$ is divisible by 2^n
- Pairing $2n$ people tells us $\frac{(2n)!}{n! \cdot 2^n}$ is an integer. So $\frac{(2n)!}{n! \cdot 2^n} \cdot 2^n$ which is what we want is also an integer.

Even Function For any function $f(k)$, $\sum_{k=0,even}^n f(k) = \sum_{k=0}^n \frac{1^k + (-1)^k}{2} f(k)$

Summation Genfunc For $a_r = b_0 + b_1 + ... + b_r$, the genfunc is $\frac{B(x)}{1-x}$

Power Series $\frac{1}{(1-x)^r} = \sum_{n=0}^\infty \binom{n+r-1}{r-1}x^n$, $1 + x + x^2 + ... + x^n = \frac{1-x^{n+1}}{1-x}$, $1 + x^n = \frac{1-x^{2n}}{1-x^n}$

Generalised Binomial Theorem $\binom{n}{r} = \frac{n(n-1)...(n-r+1)}{r!}$.

Combinatorial Identities

ID1 $\sum_{r=0}^n \binom{n}{r} = 2^n$ using $(1+1)^n$.

ID2 $\sum_{r=0}^n (-1)^r\binom{n}{r} = \binom{n}{0} - \binom{n}{1} + ... + (-1)^n\binom{n}{n} = 0$ using $(1-1)^n$.

ID3 $\binom{n}{0} + \binom{n}{2} + ... + \binom{n}{2k} + ... = \binom{n}{1} + \binom{n}{3} + ... + \binom{n}{2k+1} + ... = 2^{n-1}$ by differentiating binomial theorem.

ID4 $\sum_{r=1}^n r\binom{n}{r} = n \cdot 2^{n-1}$.

ID5 $\sum_{r=1}^n r^2\binom{n}{r} = n(n+1)2^{n-2}$ by double differentiating binomial theorem.

Example 5.8 $\sum_{r=1}^n r^3\binom{n}{r} = n^2(n+3)2^{n-3}$.

Example 5.9 $\sum_{r=0}^n (-1)^r r\binom{n}{r} = 0$.

Example 5.10 $\sum_{r=0}^{n-1}\binom{2n-1}{r} = 2^{2n-2}$.

Example 5.11 $\sum_{r=0}^n \binom{2n}{r} = 2^{2n-1} + \frac{1}{2}\binom{2n}{n}$.

Vandermonde's Identity / ID6 $\sum_{i=0}^r \binom{m}{i}\binom{n}{r-i} = \binom{m}{0}\binom{n}{r} + \binom{m}{1}\binom{n}{r-1} + ... + \binom{m}{r}\binom{n}{0} = \binom{m+n}{r}$ using $(1+x)^{m+n} = (1+x)^m(1+x)^n$.